Optimal Message Passing

Given an undirected graph $G = (V, E)$

w/ $p_e \in (0, 1) e \in E$

Find a spanning tree T^* that $\max \Pi_{e \in T}(1 - p_e)$

Minimum Spannign Tree (MST)

Given an undirected graph $G = (V, E)$

w/ $w_e, e \in E$

Find a spanning tree T' that $\min \sum_{e \in T} w_e$

MST can be solved using things like the Primm's algorithm

Reduction

Attempt to reduce the optimal message passing problem to MST

For optimization problems when trying to find a max/min, we are able to switch to a problem that looks for the min/max by changing the sign of the contents.

Reducing Optimal Message Passing problem to MST

Given a G and p for Message Passing problem, we can convert to an MST by setting the weights of our graph G' to $w_e = -\ln(1 - p_e)$

This works because $\min_T \sum_{e \in T} w_e = \min_T - \sum_{e \in T} \ln(1 - p_e) \equiv \max_T \sum_{e \in T} \ln(1 - p_e) = \max_T \ln \Pi_{e \in T}(1 - p_e) \equiv \max_T \Pi_{e \in T}(1 - p_e)$

We can do the final equivalency because the natural log is an increasing function.

T^* is opt msg passing tree iff T is MST, this is a straightforward proof since our work demonstrates all equations are either equivalent or equal.